

APPLIED MATHEMATICS: STATICS

Official Ministry of Education Textbook - Grade 12

ACADEMIC YEAR: 2025 / 2026 | GRADE 12 (GENERAL SECONDARY CERTIFICATE)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

The authorized Ministry textbook for Statics, focusing on rigid body mechanics, static friction angles, Varignon theorem of moments, ladder and hinge reactions, couples, and center of mass determination.

CORE CURRICULUM HIGHLIGHTS:

- Angle of friction and cone of limiting equilibrium
- Vector moments with cross products in 3D cartesian coordinates
- General rigid body equilibrium with ladders, hinges, and supports
- Center of gravity by negative mass subtraction and free suspension angles

APPROVED CHAPTER MODULES:

- Friction on Rough Horizontal and Inclined Planes	pp. 1 36 (2 core units)
- Moments of Forces in Two and Three Dimensions	pp. 37 72 (2 core units)
- Coplanar Parallel Forces and Beam Equilibrium	pp. 73 108 (2 core units)
- General Equilibrium of Rigid Bodies	pp. 109 140 (2 core units)
- Couples and Equivalent Systems of Forces	pp. 141 168 (2 core units)
- Center of Gravity and Composite Bodies	pp. 169 196 (2 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Official Concept & Formula Compendium (8p - 21.5 KB)
Full Ministerial Textbook: 196 Pages - 15.7 MB (Direct on MOE Portal)
Publisher: Ministry of Education & Technical Education - Arab Republic of Egypt
Official Portal: <https://moe.gov.eg/ar/elearning-content/> - Code: TH-STAT-G12
Legal Deposit: 2025/14892 - ISBN: 978-977-12-9347-78



MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

The authorized Ministry textbook for Statics, focusing on rigid body mechanics, static friction angles, Varignon theorem of moments, ladder and hinge reactions, couples, and center of mass determination. Angle of friction and cone of limiting equilibrium Vector moments with cross products in 3D cartesian coordinates General rigid body equilibrium with ladders, hinges, and supports Center of gravity by negative mass subtraction and free suspension angles

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	40% Understanding & Application Calculations, proofs & models	30% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal:

<https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Friction on Rough Horizontal and Inclined Planes	5 wks	10 pts	16.7%	2 core instructional units covering national standards and
Moments of Forces in Two and Three Dimensions	5 wks	10 pts	16.7%	2 core instructional units covering national standards and
Coplanar Parallel Forces and Beam Equilibrium	5 wks	10 pts	16.7%	2 core instructional units covering national standards and
General Equilibrium of Rigid Bodies	5 wks	10 pts	16.7%	2 core instructional units covering national standards and
Couples and Equivalent Systems of Forces	5 wks	10 pts	16.7%	2 core instructional units covering national standards and
Center of Gravity and Composite Bodies	5 wks	10 pts	16.7%	2 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: FRICTION ON ROUGH HORIZONTAL AND INCLINED PLANES

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Friction on Rough Horizontal and Inclined Planes.
- [+] Synthesize and apply analytical frameworks across all 2 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Friction on Rough Horizontal and Inclined Planes.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\left(\frac{1}{N}\right) * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: MOMENTS OF FORCES IN TWO AND THREE DIMENSIONS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Moments of Forces in Two and Three Dimensions.
- [+] Synthesize and apply analytical frameworks across all 2 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Moments of Forces in Two and Three Dimensions.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: COPLANAR PARALLEL FORCES AND BEAM EQUILIBRIUM

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Coplanar Parallel Forces and Beam Equilibrium.
- [+] Synthesize and apply analytical frameworks across all 2 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Coplanar Parallel Forces and Beam Equilibrium.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\left(\frac{1}{N}\right) * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

APPLIED MATHEMATICS: STATICS - CORE PRINCIPLES & AXIOMS
Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion: $\sum F_{net} = 0, \sum M_{pivot} = 0$
Continuity & Conservation Standard: $\text{div}(\mathbf{J}) + \partial \rho / \partial t = 0$
Normalized Standard Score: $Z = (X - \mu) / \sigma$

PROCEDURAL ASSESSMENT & TRANSFORMATION RULES
Linear Operator Transformation: $T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$
Sample Variance Boundary: $s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio: $\eta = (\text{Output_actual} / \text{Input_standard}) \cdot 100\%$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION

NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

Official Document Code: TH-STAT-G12

Curriculum Cycle: 2025/2026 Academic Year | National Edition

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